STACKS, DERIVED CATEGORIES OF SPACES: DQCOH-ALTERNATIVE-SMALL

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7596-7606 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 7608-7678, the complete author proof (69 body lines).

Editorial note. Navigation stubs: 9 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 55 of queue_spaceperfect.json (pb=3509, stmt 7596-7606, proof 7608-7678).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-DQCoh-alternative-small' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01XB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0754: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZR: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07A8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0GZN: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03DM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 071Y: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0H04: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03M1: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7678–7678. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In the situation above there are canonical exact equivalences between the following triangulated categories*

(1) $D_{QCoh}(\mathcal{O}_X)$,
(2) $D_{QCoh}(X_{affine,\acute{e}tale}, \mathcal{O}_X)$,
(3) $D_{QCoh}(X_{affine}, \mathcal{O}_X)$, *and*
(4) $QC(X_{affine}, \mathcal{O}_X)$.

Proof. If $U \to V \to X$ are étale morphisms with U and V affine, then the ring map $\mathcal{O}_X(V) \to \mathcal{O}_X(U)$ is flat. Hence the equivalence between (3) and (4) is a special case of Cohomology on Sites, Lemma 0GZR (the proof also clarifies the statement).

The discussion preceding the lemma shows that we have an equivalence of ringed topoi $(Sh(X_{affine,\acute{e}tale}), \mathcal{O}_X) \to (Sh(X_{\acute{e}tale}), \mathcal{O}_X)$ and hence an equivalence between abelian categories of modules. Since the notion of quasi-coherent modules is intrinsic (Modules on Sites, Lemma 03DM) we see that this equivalence preserves the subcategories of quasi-coherent modules. Thus we get a canonical exact equivalence between the triangulated categories in (1) and (2).

To get an exact equivalence between the triangulated categories in (2) and (3) we will apply Cohomology on Sites, Lemma 07A8 to the morphism $\epsilon : (X_{affine,\acute{e}tale}, \mathcal{O}_X) \to (X_{affine}, \mathcal{O}_X)$ above. We take $\mathcal{B} = \mathrm{Ob}(X_{affine})$ and we take $\mathcal{A} \subset PMod(X_{affine}, \mathcal{O})$ to be the full subcategory of those presheaves $\mathcal{F}$ such that $\mathcal{F}(V) \otimes_{\mathcal{O}_X(V)} \mathcal{O}_X(U) \to \mathcal{F}(U)$ is an isomorphism. Observe that by Descent on Spaces, Lemma 0H04 objects of $\mathcal{A}$ are exactly those sheaves in the étale topology which are quasi-coherent modules on $(X_{affine,\acute{e}tale}, \mathcal{O}_X)$. On the other hand, by Modules on Sites, Lemma 0GZN, the objects of $\mathcal{A}$ are exactly the quasi-coherent modules on $(X_{affine}, \mathcal{O}_X)$, i.e., in the chaotic topology. Thus if we show that Cohomology on Sites, Lemma 07A8 applies, then we do indeed get the canonical equivalence between the categories of (2) and (3) using ϵ^* and $R\epsilon_*$.

We have to verify 4 conditions:

(1) Every object of $\mathcal{A}$ is a sheaf for the étale topology. This we have seen above.
(2) $\mathcal{A}$ is a weak Serre subcategory of $Mod(X_{affine,étale}, \mathcal{O}_X)$. Above we have seen that $\mathcal{A} = QCoh(X_{affine,étale}, \mathcal{O}_X)$ and we have seen above that these, via the equivalence $Mod(X_{affine,étale}, \mathcal{O}) = Mod(X_{étale}, \mathcal{O}_X)$, correspond to the quasi-coherent modules on X. Thus the result by Properties of Spaces, Lemma 03M1 and Homology, Lemma 0754.
(3) Every object of X_{affine} has a covering in the chaotic topology whose members are elements of $\mathcal{B}$. This holds because $\mathcal{B}$ contains all objects.
(4) For every object U of X_{affine} and $\mathcal{F}$ in $\mathcal{A}$ we have $H^p_{Zar}(U, \mathcal{F}) = 0$ for $p > 0$. This holds by the vanishing of cohomology of quasi-coherent modules on affines, see discussion in Cohomology of Spaces, Section 071Y and Cohomology of Schemes, Lemma 01XB.

This finishes the proof. □